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TOPICAL REVIEW

Transformer and Large Language Models for Automatic Multiple-Choice Question Generation: A Systematic Literature Review

HALIM WILDAN AWALURAHMAN^{ID}, (Member, IEEE),
RIZAL FATHONI AJI^{ID}, (Member, IEEE), AND INDRA BUDI^{ID}, (Member, IEEE)

Faculty of Computer Science, Universitas Indonesia, Depok 16424, Indonesia

Corresponding author: Halim Wildan Awalurahman (halim.wildan@ui.ac.id)

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ABSTRACT Developing multiple-choice questions manually requires a lot of time and effort. Automatic multiple-choice question generation is one of the solutions to alleviate the problem. The research in automatic multiple-choice question generation has been growing with the recent use of Transformer and Large-Language Models. However, existing literature reviews have not thoroughly covered the recent advances in methods and evaluation conducted on the multiple-choice question generation domain. This research conducted a systematic literature review on multiple-choice question generation using Transformer and Large Language Models. This research aims to discover recent methods and evaluation strategies that have been used in the domain. We obtained 28 primary studies. We presented a taxonomy covering strategy of using the Transformer and Large Language Models for multiple-choice question generation, including fine-tuning and prompt engineering with zero-shot, few-shot, chain-of-thought, and retrieval augmented generation. Primary studies used either or both automatic and manual evaluation for the generated questions from Transformer and LLM. We found that studies are still primarily in English, with few studies utilizing learning components such as learning objective, limited use of chain-of-thought, retrieval augmented generation, and open problem in automatic evaluation.

INDEX TERMS Transformer, large language models, multiple-choice question generation, systematic literature review.

I. INTRODUCTION

Transformer and Large Language Models (LLMs) are considered to be the latest and state-of-the-art methods in Natural Language Processing (NLP). Transformer became popular from Vaswani et al. [1], which proposed a network architecture with a novel mechanism called attention mechanism. Transformer has been quickly adapted in many training data, model size, and tasks due to its performance via model pre-training, which is known as Pre-Trained Models (PTMs) [2], [3]. Transformers that were pre-trained on extensive and diverse datasets with billions of parameters are known

as large language models (LLMs) or foundation models. [4], [5]. Large Language Models excel in various tasks, including translation, text generation, data analytics, and tutoring systems [4].

The usability of LLMs and NLP, in general, have been recognized in the education domain. The particular use of LLMs in education is to generate test questions or items for assessment [6]. Developing questions are known to be difficult and time-consuming [7]. Recent research suggests that Transformer and LLMs have the capability to alleviate the challenge and assist teachers or instructors in developing usable test items [6], [8], [9]. The task is called Automatic Questions Generation (AQG). Given input in the form of text or passages, AQG will develop the question and answer

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pairs [9]. The extension of AQG in multiple-choice questions added a task called distractor generation, which aims to generate plausible distractors for multiple-choice questions format test [10].

Previous research has performed a literature review on either Automatic Question Generation (AQG), Automatic Multiple-Choice Question Generation (MCQG), or Automatic Distractor Generation (DG). Kurdi et al. [11] conducted a literature review for automatic question generation in educational background. Das et al. [12] mapped the existing datasets, modalities, and challenges in automatic question generation. Zhang et al. [9] discuss the fundamentals of the automatic question generation task, portraying many different approaches and subtasks within the automatic question generation task. Ch and Saha [7], on the other hand, discusses the automatic multiple-choice question generation process from existing literature and presents the general workflow. The recent research from Awalurahman and Budi [10] discusses automatic distractor generation as a separate task. Alhazmi et al. [13] also discussed the distractor generation task.

The existing landscape of the AQG, MCQG, and DG tasks has been explored in previous research. However, only [8], [9], [10], [13] have discussed the use of Transformers and LLM in the existing research. Transformers and LLMs have gained more attention and opportunities due to their ability and practical use. Transformer enabled a more coherent response and contextual understanding, which subsequently allowed question generation to use longer and semantically correct questions, answers, and distractors that were previously difficult to achieve [10]. LLMs, on the other hand, offer the ability to generate questions even without fine-tuning as limited MCQ training data has been problematic for Transformer approaches that use fine-tuning [14], [15]. Previous studies reported that LLMs can generate coherent and contextual responses and often surpass the performance of fine-tuned Transformer models in a wide range of tasks [4], [5]. These conditions can be beneficial for their use in MCQ generation.

While prior works have discussed the use of Transformer models and large language models (LLMs) in automatic question generation (AQG) [9] and distractor generation (DG) [10], [13], they have not provided a focused analysis of how these models are implemented explicitly in the context of multiple-choice question (MCQ) generation. Although Madri and Meruva [8] mention several Transformer-based models, they do not explore the underlying techniques in depth. In contrast, this study aims to systematically examine the specific methods and implementations of Transformer models and LLMs in automatic MCQ generation. We explored the models used and their applications, mapped the interactions between these models and fine-tuning strategies or prompt engineering techniques, and discovered what context or format has been used to utilize Transformers and LLMs in MCQ generation. Additionally, we explored how existing studies evaluate their results.

This research conducted a literature review on Transformer and LLM in automatic multiple-choice question generation tasks. The literature review was conducted using the Kitchenham framework. This research aims to discover specific implementations or methods when using Transformer and LLM in automatic multiple-choice question generation. The novelty of this research lies in its presentation of existing techniques for MCQ generation using Transformer and LLMs, which have not been explored previously. The results from this research is beneficial for the adoption of multiple-choice question generation in a new language or domain, suggesting which methods are useful or can serve as a baseline. The methods, fine-tuning strategies, prompt technique, and input context for the prompt are discussed in this study to provide future studies with directions on what to adopt and what to extend.

II. RELATED WORKS

This section discusses the previous research in AQG, MCQG, and DG. The explanations were sorted chronologically based on the earliest publications. This section aims to show how the domains of AQG, MCQG, and DG have been mapped and reviewed throughout the recent studies. The three domains often collide, in which AQG may also discuss MCQs; MCQG usually includes AQG and DG, and DG is part of MCQs. Each study has explored different objectives and research questions, yet some areas have not been thoroughly discussed.

Ch and Saha [7] was one of the first literature reviews in AQG. This study summarised the existing methods into general procedures or frameworks on AQG. The general procedures presented include pre-processing, sentence selection, key selection, question formation, distractor selection, and post-processing. For each stage, the study mentioned several different techniques. The discussion on the use of machine learning or deep learning in this study is still very limited since, in the question formation stage, a large portion of existing studies at that time used a template-based approach.

Kurdi et al. [11] provided a comprehensive review of the automatic question generation field. This research provides an overview of AQG communities and activities, summarises the existing AQG approach at that time, identifies standard evaluations of AQG performance, and tracks the changes from the previous review. This research has explained the landscape of AQG in detail by covering types of questions, types of input, domains that have been used, and overall procedures for generating questions and distractors. However, the research did not extend to more technical detail, such as mentioning the most used models or algorithms to generate questions.

Das et al. [12] extend the existing literature review by covering different modalities and assessment methods that have been used in the existing literature. This study examines the relationship between online learning assessments and various question generation strategies,

including objective, subjective, fill-in-the-blank, multiple-choice, true-false, and the integration of tables and figures into the questions. The visual question-answering format was covered in this study. Similar to [11], this research found minimal studies that used machine learning methods besides the existing NLP packages such as NLTK, SpaCy, PyNLPI, and CoreNLP because template-based QG was still popular.

Zhang et al. [9] finally covered the deep learning methods for AQG. This study covered the use of Sequence-to-Sequence (Seq2Seq) for question generation. More modern models, such as Transformers, have been integrated into this study. This study also explained different approaches when using Seq2Seq for QG, including answer-aware and answer-agnostic. Additionally, this study explained different deep learning or neural network methods in detail for QG tasks. The coverage on Transformer methods in QG has updated the domain; however, it is not the latest anymore since Large Language Models have become more and more popular [4].

Madri and Meruva [8] provided similar insight to Ch and Saha [7] but covered newer methods such as machine learning and deep learning. The general procedures of automatic multiple-choice question generation from [7] were extended in this study. The formation of questions using deep learning methods such as Recurrent Neural Network (RNN) and Transformers were mentioned. The distractor generation methods have also introduced transformer-based models such as BERT. Although the Transformer models have been acknowledged in this research, the detailed mechanism of how they were fine-tuned or inference for the MCQG task has not been explored.

Awalurahman and Budi [10] discussed the existing methods in distractor generation as a separate task of multiple-choice questions generation. This study explored the domain, types of data, types of questions, methods, and how they interact with each other within the distractor generation task. This study found that statistical methods, including transformers and deep learning models, were the most used method in distractor generation. This study still lacks an explanation of how the models were fine-tuned. The study mainly discusses ‘what’ instead of ‘how’ the models were used for distractor generation.

Alhazmi et al. [13] also explored distractor generation as an independent task. This study focused on exploring distractor generation in the fill-in-the-blank type of questions and multiple-choice questions, whereas Awalurahman and Budi [10] explored a broader range of questions, including wh-types, reading comprehension, and closest in meaning, mathematics, and visual question answering. This study has mentioned the use of Transformers and Large-Language Models, along with the strategy, including fine-tuning and prompting for distractor generation. This study was the most advanced in discussing the use of Large-Language Models in distractor generation. However, the discussion only covers the distractor generation as

an independent task, which did not include the question generation part in the multiple-choice question generation task.

Based on the previous studies, the Transformer and Large-Language Models application in automatic multiple-choice question generation has not been explored, especially in terms of how they were fine-tuned and what prompting strategies have been used. Previous studies have mentioned them several times, but the specific details of how they were implemented have not been discussed. Understanding how the models were fine-tuned and prompted could give an insight into how to adopt them in a new domain and propose an improvement based on the existing implementation. This study covered the research gap by exploring how Transformers and Large-Language Models have been prepared and employed for automatic multiple-choice question generation tasks.

III. METHODS

The study utilizes the Kitchenham [16] framework for systematic literature review. The methods consist of three stages, including planning, conducting, and reporting [10]. The planning stage involves defining the research questions, search query along with the target library, and inclusion-exclusion criteria. The conducting stage involves searching literature in databases using a defined query, filtering the results using inclusion and exclusion criteria, selecting relevant literature from titles and abstracts, and conducting a thorough quality assessment and extraction. The reporting stage presents the result of the literature review.

A. PLANNING

The planning stage was initiated by defining the scope of the literature review. In this study, only multiple choice question generation using Transformer and Large Language Models are considered. These criteria act as the scope and search query for the literature review. The research questions were defined according to the scope. The research questions defined for the literature review include the following: nousep

- 1) What are the methods in existing research?
- 2) What are the evaluation metrics that have been used in existing research?

Following the scope and research questions, the search query was defined as: “Multiple-choice questions generation” AND (“Transformer” OR “Large Language Models”). The search query was applied to four different databases, including ACM Digital Library, IEEE Xplore, Science Direct, and Scopus. The query was applied to the title, abstract, and keyword for each database except for ACM Digital Library, which used the search for all metadata instead. The inclusion and exclusion criteria were then directly applied to database results. The inclusion criteria were Journal articles and conference papers. Anything except the inclusion criteria was considered to be excluded.

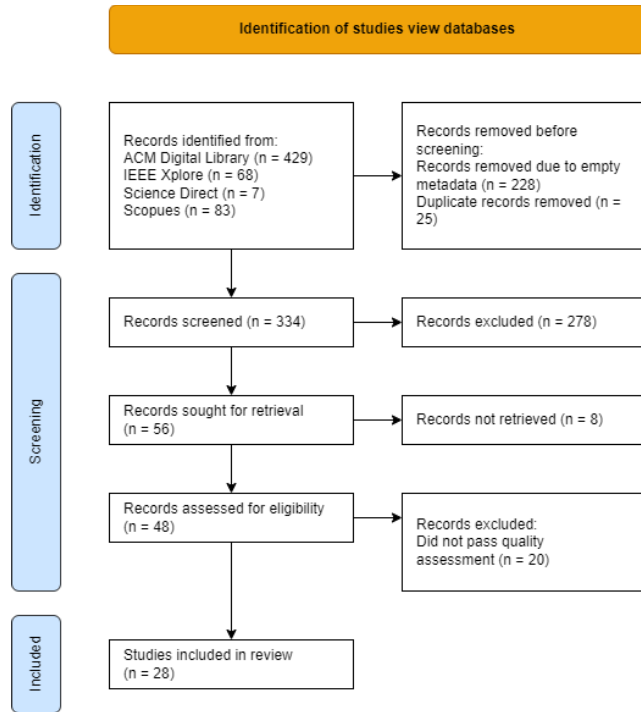


FIGURE 1. Conducting stage results.

B. CONDUCTING

The conducting stage involved searching the literature with a query until the relevant literature was used as primary sources. The summarized results of this stage are shown in Figure 1. During the screening stage, the first stage was to filter the literature based on the title and abstract. Then, the selected literature was assessed with quality assessment criteria. The quality assessment criteria were as follows: nosep

- 1) (QA1) Does the literature conduct an experiment?
- 2) (QA2) Does the literature use multiple-choice questions format?
- 3) (QA3) Does the literature use transformer or large language models?
- 4) (QA4) Does the literature explain the method?
- 5) (QA5) Does the literature mention the data used for training or testing?
- 6) (QA6) Does the literature explain the evaluation metrics?
- 7) (QA7) Does the literature attach the code?

For each question, the answer is either 1 for yes, 0.5 for partially, and 0 for no. The literature pass the assessment if the total score of the assessment is more than 4 and scored 1 in question 2.

C. REPORTING

The extraction process to answer the research questions was conducted after the primary source for the literature review was retrieved. The extraction form was defined for the

TABLE 1. Data extraction form.

No	Item	Description	Relevant RQ
1	Title		Overview
2	Authors		Overview
3	Published Date		Overview
4	Source	Journal/Conference	Overview
5	QG Model	Methods to generate question	RQ1
6	DG Model	Methods to generate distractor	RQ1
7	MCQ Model	Methods to generate complete MCQ if does not use QG-DG approach	RQ1
8	Approach detail	Detailed approach	RQ1
9	Data train	Data used for training or fine-tuning	RQ1
10	Data test	Data used for testing or inference	RQ1
11	Eval method	Manual/auto/mixed	RQ2
12	Automatic Eval Metrics	Automatic eval metric used	RQ2
13	Human Eval Metrics	Eval metrics involving human evaluator	RQ2

extraction process. The extraction form is shown in Table 1. The summary and the report for each research question are presented in Section IV.

IV. RESULTS

The results section is presented in three subsections, including an overview, research question 1, and research question 2. The overview discusses the quality assessment result and study distribution based on year and sources. Each research question subsection discussed the result relevant to the research question.

A. OVERVIEW

This study retrieved 28 works of literature as primary studies. The primary studies have passed the quality assessment stage mentioned in the method section. Figure 2a presented the result for each Quality Assessment (QA) item on primary studies. The maximum score for each QA is 28 because there are 28 papers and a maximum score of 1 per item. QA1, QA2, QA3, QA4, and QA6 scored 25 and above. Only QA5 and QA7 score less, with QA7 being the least by only 3. Based on the quality assessment results, the primary studies can be considered to answer the research questions because each QA scored well above 50% of the maximum score, with the exception of QA7.

The primary studies were published from 2021 to 2024, with most of them published as conference papers. Figure 2b presented the year and source distribution of primary studies. Only 1 study was published in 2021. Then six studies were published in 2022. The following year, 2023, saw fewer studies, with only four studies published. The latest year, 2024, however, showed an increasing number of studies,



FIGURE 2. (a) Quality assessment results and (b) year-source distribution of primary studies.

with 17 studies published, the most out of all. Regarding the source, 19 studies were published in conferences, and only nine were published in journals.

Our studies are focused on the methods since QA1, QA2, QA3, and QA4 indicated numerous studies mentioning the experiment and methods. In terms of datasets, only several datasets have been used in the primary studies. A notable trend is the frequent use of the SQuAD dataset, appearing in both training and testing phases in multiple studies such as [17], [18], [19], and [20]. Other recurring datasets include DG-RACE and RACE, used in [21], [22], [23], and [24]. Additionally, datasets such as BoolQ [17], CLOTH and RACE [24], Cosmos QA [21], EQG-RACE [22], and ScienceQA [25] appear in select works. Some papers also use domain-specific sources such as Lecture Notes [26], E-Books [27], OpenStax [14], and Wikipedia [15], [28]. Datasets such as SQuAD, DG-RACE, and BoolQ are commonly used in Transformer-based studies, where models are often fine-tuned for the MCQG task [17], [20], [23]. In contrast, datasets like Lecture Notes, Wikipedia, and ScienceQA are frequently used in LLM-based approaches, which typically apply zero-shot or few-shot prompting without fine-tuning [15], [25], [26]. In terms of code, we found only three studies shared their code, as indicated by the QA7 score. However, after further reading, we found several studies presenting their prompts in the manuscript. We do not count the prompts as code. Therefore, we omit them during quality assessment, but later, we decided to analyze them and present the results in the following subsection.

TABLE 2. MCQ models and approach.

QG Model	DG Model	Approach	Relevant Studies
mT5	mT5	fine-tune	[21]
Pegasus	T5	fine-tune	[29]
T5	Sense2Vec	fine-tune	[30, 17, 31, 19, 32]
T5	Sense2Vec	inference	[33, 27]
T5	T5	fine-tune	[23, 24]
T5	Word2Vec	fine-tune	[20]
T5	WordNet	fine-tune	[18]
GPT-3		fine-tune	[22]
Bing Chat		zero-shot	[14]
Codex		zero-shot	[34]
GPT		CoT	[35]
GPT		few-shot	[36]
GPT		one-shot	[37]
GPT		RAG	[38]
GPT		zero-shot	[26, 15, 34, 39, 40, 41]
Gemini		zero-shot	[28]
LLamaV2-70B		zero-shot	[42]
Llama 2		zero-shot	[15]
Macaw+		zero-shot	[14]
Mistral		zero-shot	[15]
QWEN-VL		CoT	[25]
Quiz Wizard		zero-shot	[39, 40]
Quizicist		zero-shot	[39, 40]

B. RQ1: WHAT ARE THE METHODS IN EXISTING RESEARCH?

The automatic multiple-choice question generation can be considered as a separate task of question generation and distractor generation or an end-to-end task consisting of a single multiple-choice question generation task [10]. Therefore, the extraction form included question generation, distractor generation, and multiple-choice question generation as separate items. Question generation and distractor generation are inseparable during extraction, which means to generate multiple-choice, question generation is always followed by distractor generation. When a study directly uses end-to-end multiple-choice question generation, larger models (LLMs) are being used and do not require other pipelines or models. The summarized results of the methods and approach are presented in Table 2.

First, we discussed the transformer-based approach. We consider studies that require separate question generation and distractor generation models to be transformer-based approaches. There are different approaches within this strategy, including Transformer-Transformer and Transformer-Lexical methods. Transformer-transformer methods used a Transformer model for both question generation and distractor generation tasks. In contrast, Transformer-Lexical methods used a Transformer for question generation and Lexical methods such as Sense2Vec, Word2Vec, and WordNet for distractor generation. The transformer-based approach can be utilized in two ways: fine-tuning and inference. Fine-tune requires re-training a model with a specific dataset. In contrast, inference does not require re-training but rather directly uses fine-tuned models available from previous studies or repositories.

The most used method in the Transformer-Lexical based approach was T5-Sense2Vec [17], [19], [30], [31], [32], which involves fine-tuning. The other two studies [27], [33] used direct inference with fine-tuned models from previous studies. The other methods have only been used once. T5-Word2Vec was demonstrated by [20] while T5-WordNet was used by [18]. Both Word2Vec and WordNet were considered older or inferior to Sense2Vec since they can only produce word-level distractor Chughtai et al. [30], Patil et al. [27]. The Sense2Vec was popular because it was trained and could discover similar words or phrases given input text [32]. Given the correct answer to the Sense2Vec model, it can generate alternative options of words/phrases that have the closest semantic similarity to the correct answer.

Some studies used the Transformer-Transformer approach. The first one was T5-T5 from [23], [24]. The second one used Pegasus-T5, which was proposed in [29]. The third one used a multilingual version of T5 called mT5, which was proposed in [21]. The T5, Pegasus, and mT5 models were all encoder-decoder models. It translates to the task of QG and DG, which often require input context that makes the encoder-decoder model suitable for sequence-to-sequence tasks such as QG and DG. Other than an encoder-decoder model, R. et al. [22] fine-tuned GPT-3, a decoder model, for multiple-choice questions generation. The Transformer-Transformer approach, however, requires fine-tuning for the models.

Second, we discussed the LLMs-based approach. We consider these approaches different from transformer-based ones due to their approach and the way previous studies utilized these models. Specifically, the LLMs approach does not require any fine-tuning. It relies on prompting strategies such as zero-shot, few-shot, chain-of-thoughts (CoT), or retrieval-augmented generation (RAG). The LLMs also have considerably larger models and parameters, such as GPT, Llama, Gemini, and Mistral.

The most used LLM is GPT. The usage of this model ranged from OpenAI API to ChatGPT. The GPT-based models have been used in various prompting strategies. Laupichler et al. [41] utilized ChatGPT 3.5 with zero-shot, Vu et al. [36] utilized GPT 3.5 with few-shot, Hang et al. [38] utilized GPT 4 with RAG. Other studies used the API version of GPT from OpenAI. Arif et al. [26] used OpenAI GPT API with zero-shot, Cheng et al. [35] used GPT API with CoT. Other studies, including [34], [39], [40] used the ChatGPT version of the GPT model. Besides GPT, other models such as Bing Chat, Codex, Gemini, Llama, Macaw, Mistral, and Qwen have also been utilized in MCQ generation studies, mostly using zero-shot approach [14], [15], [25], [34], [42].

Third, we discussed the variety of fine-tuning formats or input-output structures for fine-tuning and prompting strategies that have been utilized in previous studies. The fine-tuning approach has mainly been used in the question generation task. The distractor generation tasks that require fine-tuning were only those that used Transformer models. The pre-trained models, such as Sense2Vec, Word2Vec,

and WordNet, did not require fine-tuning for the distractor generation task [18], [20], [32]. There are two distinguishable formats for the fine-tuning scheme: highlighting and prepending the input. For the prompting strategies, various formats and combinations can be used as an example. The main difference between each approach can be distinguished. For instance, zero-shot consists of direct instruction, one-shot or few-shot gives an example of the intended input-output for the model, and CoT requires structured and sequential prompts to feed to the model. In contrast, RAG requires additional information from external sources. We presented the examples in Table 3.

A *prompt* analysis was conducted to find the format and input required in the *prompt*. The input into the *prompt* was used as the context of [4]. Prompt analysis was done manually. The context entered into the prompt is usually marked with “{ “ ”}” or “[“ ”]”. The content entered into the *prompt* is first determined or created by the user. Therefore, the user must first prepare the content. To make it easier to explain, the content in this study is called features/input. Not all previous studies were used in the analysis because only some of them display *prompt*. Based on the *prompt* analysis, the most used feature in previous studies was context in the form of text/paragraph, which was six times. The topic was the second most used feature, with five uses. The subject was the third feature with three uses. Learning objectives, number of questions, and type of questions followed with two uses. Meanwhile, answers, domains, speaking style/tones, files, and Bloom’s taxonomy were only used once. These results do not include studies that did not mention the *prompt* used. However, it can be used as an illustration that previous studies have used a lot of text as input to be processed into multiple-choice questions, similar to the *sequence-to-sequence* approach in the Transformer model. Some interesting attributes to use but have not been widely adopted are Bloom taxonomy levels, learning objectives, and files with the RAG scheme. The combination of the input and overall results are shown in Figure 3.

C. RQ2: WHAT ARE THE EVALUATION METRICS THAT HAVE BEEN USED IN EXISTING RESEARCH?

The evaluation for automatic multiple-choice question generation has various approaches or methods. We consider three types of methods: automatic, manual, and mixed. Automatic evaluation is an evaluation using metrics in which, during the process, it does not require human intervention. For example, measuring the word overlap of generated questions or distractors to the reference directly with metrics such as ROUGE or BLEU. Manual evaluation is an evaluation that requires human intervention. For example, a human annotator can assess fluency and difficulty, or use a rubric like item writing flaws. Mixed evaluation, therefore, is the combination between the two. The findings for the evaluation methods on primary studies are presented in Table 4.

TABLE 3. Fine-tuning and prompting strategies examples.

Approach	Example
Fine-tuning QG	1) context: (-summary-) answer: (-keyword-) </s> [27] 2) generate question: context <hl> answer <hl> context [17]
Fine-tuning QG-DG zero-shot	generate question: context <hl> answer <hl> context generate distractor: question answer: answer context: context [23] 1) Write 10 MCQs about: ... [39, 40] 2) Write a multiple choice question using the following sentence and answer. Convert the sentence into a question that matches the answer. Use JSON format. Sentence: <sentence> Answer: <answer> [14]
few-shot	Role: You are a lecturer of [subject name] Task: Please create [a specific number] questions Context: [The kind of exercise] questions that focus on the content of [name of the lesson to test] Example: (This is the example I want you to imitate: [specific example of the type of exercise to create]) Format: Present output in [format name] [36]
chain-of-thoughts	Stage 1: Generate concrete rules for writing the distractors in multiple-choice questions regarding the statement "MD5 is an example of a hashing algorithm". The rules for creating the distractors should be based on the subject, predicate, and object presented in the statement. The statement might appear in the form of the following question template. [object] + is + [concept]. For example, [object] + is + [example 1] + [example 2] + [example 3]. Stage 2: Generate several distractors regarding the statement "MD5 is an example of hashing algorithm". The distractors should be generated based on the following rules. <Output from previous stage: 1. Ensure the distractors are other hashing algorithms. 2. Include distractors that are not related to hashing algorithms. 3. Create distractors that are incorrect or false statements about MD5.> [35]
retrieval-augmented	Create 10 multiple choice questions for the provided file. Each question shall have 4 answers and only 1 correct answer. The output shall be in JSON format, i.e., an array of objects where each object contains the stem, an array for the answers and the index of the correct answer. Name the keys 'stem', 'answers', 'correctAnswerIndex'. The output shall only contain the JSON, nothing else. [40]

TABLE 4. Type of evaluation and the number of studies utilized them.

Method	No. of studies	Relevant Studies
Auto	7	[35, 21, 17, 42, 31, 18, 28]
Manual	12	[15, 30, 38, 37, 34, 39, 40, 33, 41, 14, 27, 36]
Mixed	9	[26, 32, 22, 29, 25, 23, 19, 24, 20]

The automatic evaluation was used less than the other methods, with only seven relevant studies. The manual evaluation was the most used approach, with 12 relevant studies. The mixed evaluation has been used in 9 other studies. Many studies used a manual approach because generating high-quality items is very important. Most studies rely on experts to judge the quality of the generated multiple-choice questions. The existing automatic evaluation mostly only measured the overlap of the generated item with the reference, which makes it very limited. Therefore, to provide a better understanding of the results, studies used both approaches.

We found several automated evaluation metrics presented in Table 5. Some of the metrics were used only in one of the QG or DG tasks, while the others can be used in both tasks. BLEU and ROUGE metrics were the most used ones

TABLE 5. Automatic evaluation metrics and their respective task.

Auto Eval Metric	QG	DG	Relevant Studies
BLEU	x	x	[21, 22, 17, 25, 23, 24, 20, 25]
Exact Matches	x	x	[18]
F1-Score	x	x	[17, 19]
Fischer Information	x	x	[42]
Item Response Theory	x	x	[42]
Item Writing Flaws	x	x	[26]
Meteor	x	x	[17]
Precision	x		[19]
Recall	x		[19]
ROUGE	x	x	[21, 32, 22, 17, 31, 29, 25, 23, 20]
Semantic Similarity	x		[24, 28]
Student Score	x	x	[35]

and can be used for both QG and DG. Other metrics such as F1-score, METEOR, and similarity have been used by two studies each. The other metrics have only been used once. Some metrics still involve participants, such as Item Response Theory [42] and student score [35]. However, we consider them as automated metrics since the calculation was done using established tools and does not require expert judgment. For example, [42] used the GPT model as a student

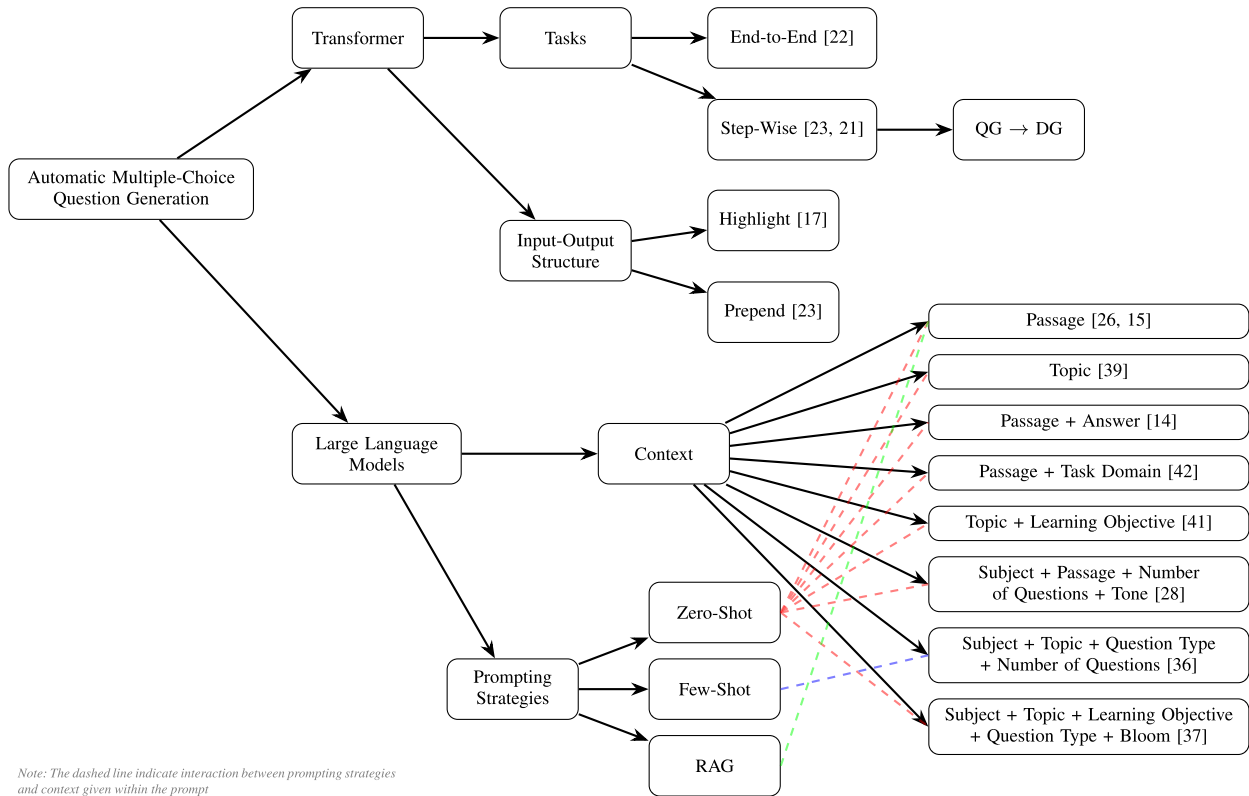


FIGURE 3. Taxonomy of automatic multiple-choice question generation using transformers and LLMs.

to directly evaluate the questions generated. The Item Writing Flaws metric could also be used as a manual evaluation since it consists of many criteria that experts can check. However, Arif et al. [26] used a rule-based tool to automatically tag the flaws within the generated questions, therefore making it automated.

We also found several metrics involving expert or human participants as presented in Table 6. There were many variations and types of metrics in the manual approach. Some studies used relevance, fluency, grammar, and difficulty. The presented table showed only a few of the most used metrics. Many other metrics have been used throughout primary studies. However, we only selected metrics that are similarly used across the study to be shown in Table 6. There was also a more analytical approach called docimology by [39] and [40], which has different dimensions of evaluation. We consider it an interesting approach that can be used for a more detailed examination of generated MCQs.

V. DISCUSSION

The results from this study added more insight into previous literature studies. This study highlights the use of Transformer and LLMs with fine-tuning and prompt engineering approach, which differs from Ch and Saha [7] that summarized procedures and tools for MCQ generation and Kurdi et al. [11] that are concerned about the general picture of AQG in terms of data, tools, and evaluation

metrics. This study also differs from Das et al. [12] because we only focused on one type of question, multiple-choice, and highlighted the use of more recent technologies, including Transformer and LLMs. Zhang et al. [9] provided deep learning and Transformer methods in QG but did not explore the distractor and MCQ. This study extends Zhang et al. [9] by exploring LLMs and using MCQ instead of the regular question-and-answer format. Madri and Meruva [8] and Awalurahman and Budi [10] did not explore technical perspectives like fine-tuning format and prompt engineering technique, including the context needed as input, which is now discussed in this study. Finally, Alhazmi et al. [13] explored LLMs for distractor generation, so we extend their insight by providing DG and QG as unified tasks in MCQ generation. The following subsection discusses existing research methods, evaluation methods, limitations, and future works that we found during our study.

A. RQ1: WHAT ARE THE METHODS IN EXISTING RESEARCH?

There were two approaches to develop the MCQ generation model or pipeline. First, use the combination of the QG model and the DG model as separate tasks. Second, use one model directly for QG and DG as a single task. We found that most primary studies used a single-task approach. The result differs from [10], which stated that the distractor generator

TABLE 6. Manual evaluation metrics and their description.

Manual Eval Metric	Description	Relevant Studies
Answerability	Whether the question can be answered using the provided context	[26, 38, 22, 19]
Appropriateness	Whether the question aligned with the subject or exams	[34, 25]
Clarity	Whether the meaning of the question is clear and understandable	[26, 15, 20]
Coherence	whether the answer and distractors are relevant to the question	[15, 22]
Complexity	Whether the questions require higher thinking skills to answer	[38, 25]
Difficulty	How difficult is it to answer the questions	[26, 34, 40, 33, 23, 19, 41]
Distinctiveness	Whether the distractors are different from one to another	[25, 14]
Docimology Analysis	Specific criteria list used to evaluate the question	[39, 40]
Fluency	Whether the question and options are grammatical and fluent	[22, 14, 19, 20]
Grammar	Whether the question are consistent grammatically	[26, 38, 39, 40, 19]
Plausability	Whether the distractors are convincing and likely to be selected	[26, 25, 14]
Quality	How well does the questions is structured or formed	[14, 23]
Relevancy	Whether the question is relevant to the subject or provided context	[26, 38, 32, 22, 33, 14, 27, 19, 20]

has been increasing, as we focus on studies that used a Transformer in one or both tasks, whereas they did not apply this scope. The model used for the two branches of approach also differ. Based on our findings, the first approach mostly utilized smaller Transformer based models such as mT5, Pegasus, or T5 [21], [23], [29] for QG model and variations of Sense2Vec [17], [30], Word2Vec [20], WordNet [18], or the same Transformer model [21], [23]. The second approach used bigger model such as GPT [34], Gemini [28], Llama [15], Mistral [15], or QWEN [25].

The difference in models affects how the primary studies utilize them. There were also two branches of approach in this matter. First, studies that used separate QG-DG tasks mostly used a fine-tuning approach. Second, studies that used QG-DG as a single task mostly used a prompting approach. The individual task requires fine-tuning due to the model's capability and its counterpart in the DG task. The study that used Sense2Vec, Word2Vec, and WordNet for DG did not fine-tune the DG model but only fine-tuned the Transformer QG model for QG [32]. The prompting approach can be used directly to the model without fine-tuning.

Different models also resulted in different MCQ outputs. For studies that used Transformer-Sense2Vec models, the distractors generated were in word or phrase level [30]. The Sense2Vec model has inherent limitations that struggle to create coherent sentences. Therefore, the distractors generated were shorter. The same condition applies to Word2Vec and WordNet. The Transformer-Transformer models can produce sentence-level distractors, which are typically longer, making them suitable for reading comprehension types of MCQs [21], [23]. For studies that used LLMs, the MCQ results were considered better, but there are still issues following the generated MCQs. The problems include very long answer keys, ambiguous answer keys, implausible distractors, or even no answer keys available [40].

The fine-tuning and prompting approaches have different strategies. We took a look at the example given by the primary studies. For QG fine-tuning, there were prepend [27] and highlight strategy [17]. Prepend separated the context

and answer to generate questions while highlighting directly adding `<h1>` tag within the context to indicate the answer span. For QG-DG fine-tuning, the prepend and highlight strategies were still used. Additionally, there was a strategy to fine-tune the model to generate the wrong options one by one [24]. For prompting approach, there were zero-shot [39], [40], few-shot [36], chain-of-thoughts [35], and retrieval-augmented generation strategy. Every study used different wording or prompts but still resembled the characteristics of each strategy. We compared each strategy and presented the results in Table 7.

Furthermore, using LLMs offers the possibility of integrating more context to the question generation task beyond just a paragraph. Doughty et al. [37] demonstrated that it is possible to incorporate learning objectives for the context and let LLMs write the MCQs based on given learning objectives. Additionally, LLMs offer more flexibility in terms of the type of questions, depending on the model capability, which includes visual MCQs [25]. There is a reason why our findings in Table 2 indicated that there are more studies that utilize LLMs compared to Transformers. The flexibility and broader range of text-generation ability from LLMs remain the most significant advantage over the previous Transformer models. We compared the components or inputs that can be used for the prompt along with their pros and cons in Table 8.

Fine-tuning and prompting are two paradigms that offer different benefits and drawbacks. Fine-tuning is typically used for domain-specific data or tasks, which allows the model to specialize in the training objective. However, fine-tuning requires a large amount of data. For MCQ generation, it requires an MCQ dataset. There might be existing datasets such as RACE or SQuAD. Still, the dataset is limited to its current version, making fine-tuning it less varied and limited to what the datasets offer. Therefore, fine-tuning may only be suitable if we have a specific domain dataset for one particular task. Prompting, on the other hand, does not require re-training the model. Instead, it relies on how the user formulates the prompt to utilize the model to its full capability. Many different prompting techniques have been

TABLE 7. Comparison of combination of methods for MCQ generation.

Method	Detail	Description	Pros	Cons
Fine-Tuning	End-to-End	Question, answer key, and distractor in one-go	Produce the MCQ at once; more efficient in terms of fine-tuning (only 1 model is needed) [22]	Valid answer and plausible distractors are still challenging [22]
Fine-Tuning	Step-Wise	Question produced first, then answer, then distractors	Allows for more control over the quality of each question, answer, and distractor as they are sequential. [22, 23]	Requires fine tuning on 3 different tasks [23]. If the model does not support multi-tasking, it will require 3 models. [22]
Fine-Tuning	Highlight Format	Formatting the input by tagging relevant phrases. e.g. <h1>answer<h1>	Helps the model learn by indicating which tokens are more important [17, 21]	Over reliance on the presence of tags.
Fine-Tuning	Prepend Format	Formatting the input by adding components after another.	More readable; allows model to adapt even when there is no hint or tags present. [23]	Susceptible to irrelevant answers or distractors over questions [22]
Prompting	Zero-shot	Only instruction is given to the model	More direct	No additional context; Reliance on models internal knowledge [40]
Prompting	Few-shot	Instruction and example of response are given to the model	Model can learn and imitate the example response [37, 36]	Has to find and select relevant example. [36]

TABLE 8. Comparison for each component used within the prompt.

Component	Description	Pros	Cons
Passage	Paragraph or text given as the source of the question.	Questions are directly inherited from source text [26]	Creating or finding passages still requires effort [15]
Topic	Main discourse, or ideas being discussed.	-	Mostly depends on model's internal knowledge [40]
Answer	Target answer for the question, usually accompanied by the passage.	Questions are oriented from context and answers [14]	Requires effort to find or formulate the answer [14]; question generation models sometimes requires answer extraction stage [17]
Learning Objective	The aim of the question in regards of learning objective. i.e. What competencies, skills, or knowledge are being tested using the question.	Questions are directed for specific learning objectives. [41, 37]	Learning objective requires formulation according to the course target [37, 41]; The questions need to be validated for relevance [37]
Subject	The subject or class in education level	Broader domain compared to topic[36]	Mostly depends on model's internal knowledge
Number of Questions	Amount of questions needed to be made.	-	-
Tone	Overall sound or writing styles used for the question.	-	-
Question Type	The type of questions needed for the assessment. It usually depends on the subject and technical definition.	Gives clear direction and specification for the question structure or content [37]	Has to be defined first according to the subject/course [37]
Bloom	Cognitive level based on Bloom taxonomy.	Provide different level of questions; Can be helpful to measure the learning objectives [37]	Requires model to understand the context and mapping between cognitive level, questions, and content of the course.

popularized. However, the single most significant drawback of using prompting and LLMs is that it requires a tremendous amount of computing power. LLMs are resource-heavy, and the more parameters they have, the more memories and computing power they consume. That is, if the LLMs are hosted independently using the user server.

B. RQ2: WHAT ARE THE EVALUATION METRICS THAT HAVE BEEN USED IN EXISTING RESEARCH?

There are three different types of evaluation approaches: automatic, manual, and mixed. The manual approach was the most used one. Comparing automatic evaluation metrics and manual evaluation metrics revealed that automatic

evaluation metrics primarily relied on word overlap to measure performance. The automatic evaluation metrics may be used for one or both of the QG-DG tasks. In contrast, manual evaluation metrics consider the MCQ as a whole. The manual evaluation metrics rely on expert judgments, thus making the evaluation more appropriate and closely resembles the manual question generation task. Additionally, with a manual evaluation approach, the quality of the questions can be observed and maintained.

The automatic evaluation metrics from primary studies mainly focused on measuring overlap words from reference and generated items. There are well-known metrics such as BLEU, ROUGE, and METEOR, which, if simplified, all measure how similar one text is to another based on the

overlap. Other standard metrics for machine learning include recall, precision, f1-score, and exact matches. These metrics also measured how similar the generated item is with the reference.

Previous studies have highlighted the lack of a standardized approach to evaluate the MCQ automatically. A study from [42] utilized another LLM to act as students to assess the MCQ using Item Response Theory, making the process semi-automated. The previously requiring human participants metrics can now be substituted using LLMs. Thus reducing the effort for the manual evaluation. However, using the model still requires considerable computational power. Another semi-automated approach was used by Arif et al. [26], which utilized item writing flaws in a rule-based format by automatically tagging the flaws within the MCQs.

The manual approaches that have been used in primary studies were distinct from one another. The terms used might be slightly different, but the intention might be similar. Grévisse et al. [40] proposed a thorough analysis through Docimological analysis. The evaluation proposed within the Docimological analysis covers many aspects of writing MCQs. If future studies were to be evaluated manually, considering the previous studies' metrics would be useful and help establish the baseline.

C. LIMITATION

This study has several limitations. First, because the authors created and carried out the quality evaluation for this study, there may have been some subjectivity added that affected the findings. It is advantageous to consider including external evaluation in the quality assessment procedure in further research. Second, this study specifically chose Transformer and Large-Language Models for the automatic multiple-choice question generation task. There are various methods other than Transformer and LLMs, which can be discussed in future studies. Third, this study has contrasted the difference between each method and strategy. However, comparative studies need to be conducted to decide which methods performed the best. The existing discussion was based on the primary studies' findings and recommendations. Currently, some of the existing methods use completely different input and output formats from the others while also using different evaluation metrics. Therefore, future studies aiming for performance comparison should develop unified testing for this variety of use cases and formats.

D. FUTURE WORKS

We found several avenue for future studies based on the findings:

- 1) The Transformer-Sense2Vec approach has been used several times, but they all used English. There were no studies that have adopted this approach in other languages. Additionally, the on-the-go version of Sense2Vec was only trained in English. Therefore, this approach may not be replicated in different languages and needs some interventions. The potential approach would be to use another representation model, such as FastText.
- 2) The fine-tuning approach requires a large-scale dataset. The commonly used datasets were RACE and SQuAD. Both datasets mainly focused on reading comprehension. Although RACE has distinguished the level of questions, the widely used reference, such as Bloom's taxonomy, has not been integrated into these datasets. Thus, the relations between generated questions and bloom level have rarely been explored. The few studies that have explored Bloom and Learning Objectives were [37], [41], [42], all of them utilized LLMs instead of fine-tuning Transformer models. There might be inherent limitations on Transformer models for this kind of question, but it needs to be explored further.
- 3) Many studies used zero-shot, but other variations, such as few-shot, CoT, and RAG, were still limited.
- 4) Most prompts were written in English and intended to be used for English MCQs. There is unexplored field for other languages MCQ generation using LLMs. C. [39] proposed an intriguing idea: exploring how an English prompt performs when the target language is defined for other languages, such as using an English prompt but asking the model to generate output in Indonesian. Besides using the same language for prompt and output, exploring multilingual prompt-output could be a future research direction.
- 5) For the evaluation methods, since most studies used manual evaluation, the automation of manual metrics remains an open research. Specifically, how to leverage NLP techniques to judge the quality of the generated MCQs. For example, using LLM for manual evaluation G. et al. [42].

VI. CONCLUSION

This study has explored how Transformer and LLMs are being used in automatic multiple-choice question generation tasks. Previous research has discussed the use of Transformer and LLM models for automated question generation and distraction, but none has specifically analyzed their implementation in multiple-choice question (MCQ) generation. This study systematically examines the methods, applications, and evaluation strategies used in utilizing Transformer and LLM for MCQ generation, including fine-tuning techniques and prompt engineering. Transformer models are being used in multi-task MCQG where question and distractor are developed separately. Transformer models can be used in pairs with lexical methods such as Sense2Vec, Word2Vec, and WordNet or other Transformer models. The Transformer models need to be fine-tuned for MCQG. The LLMs are being used as a single task MCQ where question and distractor are generated at once by the same model. The LLMs are being utilized with prompting strategies, including zero-shot,

few-shot, chain-of-thoughts, and retrieval-augmented generation. For future studies, we suggest exploring the use of the lexical model in other languages, more variations on prompting strategies, and multilingual prompt strategy.

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HALIM WILDAN AWALURAHMAN (Member, IEEE) received the B.Sc. degree in computer science from Universitas Airlangga, Surabaya, Indonesia, in 2023, and the M.Sc. degree in computer science from Universitas Indonesia, Depok, Indonesia, in 2025, where he is currently pursuing the Ph.D. degree in computer science, under the Pendidikan Magister menuju Doktor untuk Sarjana Unggul (PMDSU) Scholarship from the Ministry of Research, Technology, and Higher Education, Republic of Indonesia. His research interests include natural language processing and software engineering. He is also researching automatic question generation in Bahasa Indonesia.



RIZAL FATHONI AJI (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from Universitas Indonesia, Depok, Indonesia. He is a Lecturer and a Researcher with the Faculty of Computer Science, Universitas Indonesia. He is currently the Manager of Information Technology at the Faculty of Computer Science. He is also a member with the Computer Systems Laboratory, Center for Cyber Security and Cryptography, and the Digital Library and Distance Learning Laboratory, Universitas Indonesia. His research interests include computer network and information security with expertise in cyber security, infrastructure and cloud computing, and e-learning.



INDRA BUDI (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in computer science from Universitas Indonesia, Depok, Indonesia. He is currently a Professor with the Faculty of Computer Science, Universitas Indonesia. His research interests include natural language processing, information extraction, social media analytics, sentiment analysis, and text mining. He has contributed extensively to research on social media sentiment during elections, hate speech detection, and fintech analysis. He serves as an assessor for the National Accreditation Board for Higher Education (BAN-PT) and the Informatics and Computer Independent Accreditation Agency (LAM INFOKOM), Indonesia.

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